Carson S. Britt

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Profile

Chemistry-trained ML researcher who designs and trains models for chemical reaction tasks and scientific data extraction, including a pipeline that extracted and structured 13M+ academic literature and patent passages at a cost compatible with bootstrapping a startup. Cofounder of De Novo Chem, which uses that extraction pipeline to power Saguaro Chem, a searchable chemical reaction database, at denovochem.com

Experience

De Novo Chem - Cofounder
Glendale, AZ — April 2022 - Present

- Designed and built an LLM fine-tuning pipeline for chemistry information extraction: manually labeled a dataset for reaction extraction tasks, fine-tuned Qwen-family models with QLoRA.
- Built custom Python/vLLM infrastructure for large-scale distributed inference, handling tens of millions of extraction requests with automated batching and output format validation.
- Architected and built Saguaro Chem, a chemistry search platform indexing 6M+ compounds, 13M+ reaction passages, and 15M+ characterization details, searchable by structure, substructure, similarity, and references, with a Next.js/FastAPI/Python/PostgreSQL-RDKit stack.
- Developed Cholla Chem, Agave Chem, and rdchiral_plus, open-source Python cheminformatics libraries for chemical name-to-SMILES resolution, reaction atom-mapping and classification, and retrosynthetic template extraction and application, respectively.

CS Squared LLC - Research Scientist
Glendale, AZ — January 2019 - Present

- Forked and modified AiZynthFinder, improving search speed by 5-10x, implementing custom neural template prioritization models, reaction feasibility models, and reaction condition models.
- Developed and deployed 23 XGBoost models for molecular property and toxicity prediction using RDKit descriptors; validated with nested 5-fold cross-validation and benchmarked as competitive with published SOTA for these datasets. Integrated into Schrödinger LiveDesign.
- Set up, administered, and deployed computational chemistry, machine learning workflows, and custom retrosynthesis tool in Schrödinger LiveDesign.

Education

Case Western Reserve University, Cleveland, OH — May 2018
B.S. in Chemistry, Minors in Mathematics and German

- Earlier research: synthetic/polymer chemistry at Cornell and CWRU (2015–2018), including air-free synthesis and characterization techniques (NMR, MALDI-TOF, cyclic voltammetry).

Technical Skills

- **Machine Learning:** PyTorch, TensorFlow, XGBoost, LLM fine-tuning (QLoRA), vLLM, semi-supervised learning, synthetic data generation, hyperparameter optimization
- **Chemistry / ML:** RDKit, cheminformatics, retrosynthesis, computer-aided synthesis planning, reaction extraction, molecular structure, similarity, and substructure search
- **Programming:** Python, TypeScript, JavaScript, SQL
- **Software / Systems:** FastAPI, Next.js, PostgreSQL, Redis, REST APIs, Docker, Git, CI/CD
- **Infrastructure:** GPU inference, distributed/batched inference, RunPod, Railway